Calculo ~~Integral~~ 6/8

7. $f(x)=\sqrt[3]{x^2}$

13. $f(x)=\dfrac{2x^4-3x+1}{x+1}$

21. $f(x)=x^2 \operatorname{sen} x$

22. $f(x)=(x^2+1)\operatorname{Tan} x$

25. $f(x)=\dfrac{\operatorname{sen} x}{x^2+1}$

28. $f(x)=3xe^{3x+5}$

29. $f(x)=\ln(x^2)-e^{5x}$

30. $f(x)=e^{2x}.\operatorname{sen}(x)$

31. $f(x)=\operatorname{sen}(x^2-3x)$

33. $f(x)=\sqrt[3]{(3x^2-5)^2}$

34. $f(x)=\dfrac{\operatorname{sen}(x+2)}{3x^2}$

7. $f(x)=\sqrt[3]{x^2} \Rightarrow f'(x)=\dfrac{2x^{-\frac{1}{3}}}{3}$

$= x^{\frac{2}{3}}$ $\qquad = \dfrac{2}{3\sqrt[3]{x}}$

$f(x)=2^{2x} \Rightarrow f\circ g \Rightarrow 2^x$

$g \Rightarrow 2x$

13. $f(x)=\dfrac{2x^4}{x+1}-\dfrac{3x}{x+1}+\dfrac{1}{x+1}$

$\Rightarrow f'(x)=\dfrac{8x^3(x+1)-2x^4\cancel{(x+1)^2}}{(x+1)^2}-\dfrac{3(x+1)-3x\cancel{(x+1)^2}}{(x+1)^2}+\dfrac{-\cancel{(x+1)^2}}{(x+1)^2}$ $\quad 1$

$= \dfrac{8x^3(x+1)-2x^4-((3x+1)-3x)-1}{(x+1)^2}$

$\dfrac{\operatorname{sen}(x)}{x^2}$

21. $f(x)=x^2\operatorname{sen} x \Rightarrow f'(x)=2x\operatorname{sen}(x)+x^2\cos(x)$

22. $f(x)=(x^2+1)\operatorname{Tan}(x) \Rightarrow f'(x)=2x(\operatorname{Tan}(x))+(x^2+1)\sec^2(x)$

25. $f(x)=\dfrac{\operatorname{sen}(x)}{x^2+1} \Rightarrow f'(x)=\dfrac{\operatorname{sen}'(x)(x^2+1)-\operatorname{sen}(x)(x^2+1)'}{(x^2+1)^2}$

$=\dfrac{\cos(x)(x^2+1)-2x(\operatorname{sen}(x))}{(x^2+1)^2}$

28. $f(x)=3xe^{3x+5}$

$\Rightarrow f'(x)=\cancel{3x\cdot e^{3x+5}}=3e^{3x+5}+3x\cdot 3e^{3x+5}=3e^{3x+5}+9xe^{3x+5}$

$=3e^{3x+5}(1+3x)$

29. $f(x)=\ln(x^2)-e^{5x} \Rightarrow f'(x)=\dfrac{1}{x^2}.2x-e^{5x}.5=\dfrac{2x}{x^2}-5e^{5x}$

30. $f(x)=e^{2x}.\operatorname{sen}(x) \Rightarrow f'(x)=2e^{2x}\operatorname{sen}(x)+e^{2x}\cos(x)$

31. $f(x)=\operatorname{sen}(x^2-3x)$

$f'(x)=\cos(x^2-3x)\cdot(2x-3)$

33. $f(x)=(3x^2-5)^{\frac{2}{3}} \Rightarrow f'(x)=\frac{2}{3}(3x^2-5)^{-\frac{1}{3}}.(6x)=\dfrac{12x}{3\sqrt[3]{(3x^2-5)}}$

34. $f(x)=\dfrac{\operatorname{sen}(x+2)}{3x^2} \Rightarrow f'(x)=\dfrac{\operatorname{sen}'(x+2)(3x^2)+-\operatorname{sen}(x+2)(3x^2)'}{(3x^2)^2}=\dfrac{3x^2\cos(\cancel{x+2})-\operatorname{sen}(x+2)6x}{9x^4}$

Cal Int 5/8 Prof. Horacio

Propiedades de la derivada: Sean $f, g : [a,b] \to \mathbb{R}$ funciones derivables y $\alpha \in \mathbb{R}$

i) $(\alpha f(x))' = \alpha f'(x)$

ii) $(f(x) \pm g(x))' = f'(x) \pm g'(x)$

iii) $(f(x) g(x))' = f'(x) g(x) + f(x) g'(x)$

$(fgh)' = f'gh + fg'h + fgh'$

iv) $\left(f(x)\frac{1}{g(x)}\right)' = \left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}, \quad g(x) \neq 0$

$f(x) = \frac{2x^4 - 3x + 1}{x} \Rightarrow f'(x) = 6x^2 + -x^{-2}$

$f(x) = \frac{2x^4}{x} - \frac{3x}{x} + \frac{1}{x}$

$= 2x^3 - 3 + \frac{1}{x}$

$f'(x) = \frac{(8x^3 - 3)\cdot x - (2x^4 - 3x + 1)}{x^2}$

$= \frac{8x^4 - \cancel{3x} - 2x^4 \cancel{+3x} - 1}{x^2}$

$= \frac{6x^4 - 1}{x^2}$

$= 6x^2 - \frac{1}{x^2}$

b) $f(x) = \frac{(2x^2+1)\,\text{sen}(x)}{x^2+1}$

$f'(x) = \frac{(4x(\text{sen}(x)) + (2x^2+1)\cos(x))(x^2+1) - (2x^2+1)\text{sen}(\ldots}{(x^2+1)^2}$

$= \frac{[4x\,\text{sen}(x) + (2x^2+1)\cos(x)](x^2+1) - (4x^3+2x)\text{sen}(x\ldots}{(x^2+1)^2}$

c) $f(x) = (x^2-1)\sec x \tan x \Rightarrow f'(x) = (x^2-1)'\sec x \tan x + (x^2-1)\sec' x \tan x + (x^2-1)\sec x \tan' x$

$f'(x) = 2x \sec x \tan x + x^2 - 1 \sec x \tan^2 x + (x^2-1)\sec^3 x$

Si $(g \circ f)(x) = g(f(x))$

Regla de la cadena $\rightarrow$ $(g \circ f)'(x) = g'(f(x)) \cdot f'(x)$

Ej: $f(x) = \text{sen}(2x^2+1) \Rightarrow f'(x) = \cos(2x^2+1)\cdot 4x = 4x\cos(2x^2+1)$

Calculo Integral Prof. (H)Oracio Sambrueza. 9 v/s Total [6 controles]

Echo de clase < 75% asistencia

"examen"

NP $\geq 4,95$ para eximirse de la evaluación "examen"

Solemne recuperativa en caso de faltar a una prueba [de carácter global]

Unidad 0: Repaso de derivadas.

Derivada → la pendiente de la recta Tangente a una curva en un punto dado.

$$m = \frac{f(x+h) - f(x)}{h}$$

Si $h \to 0$, se obtiene la pendiente de la recta tangente en el punto X.

$$\lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = f'(x) = \frac{dy}{dx} = \frac{df(x)}{dx}$$

3B GRAPHITE ARTEL

Ejemplo. $f(x) = \sqrt{x}$

$f(x+h) = \sqrt{x+h}$

$$f'(x) = \lim_{h \to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}$$

$$= \lim_{h \to 0} \frac{(\sqrt{x+h} - \sqrt{x})(\sqrt{x+h} + \sqrt{x})}{h(\sqrt{x+h} + \sqrt{x})} = \lim_{h \to 0} \frac{(x+h-x)}{h(\sqrt{x+h}) + h\sqrt{x}}$$

$$= \lim_{h \to 0} \frac{\cancel{h}}{\cancel{h}(\sqrt{x+h} + \sqrt{x})}$$

$$= \lim_{h \to 0} \frac{1}{\sqrt{x+h} + \sqrt{x}}$$

$$= \frac{1}{2\sqrt{x}}$$

$f(x) = x^{\frac{1}{2}}$

$$f'(x) = \frac{1}{2} \cdot x^{\frac{1}{2}-1}$$

$$= \frac{1}{2} \cdot x^{-\frac{1}{2}}$$

$$= \frac{1}{2} \frac{1}{x^{\frac{1}{2}}} = \frac{1}{2\sqrt{x}}$$

Tabla de derivadas Trigonométricas.

$f(x) = sen(x) \Rightarrow f'(x) = cos(x)$

$f(x) = cos(x) \Rightarrow f'(x) = -sen(x)$

$f(x) = Tan(x) \Rightarrow f'(x) = sec^2(x)$

$f(x) = sec(x) \Rightarrow f'(x) = sec(x) Tan(x)$

$f(x) = csc(x) \Rightarrow f'(x) = -csc(x) cot(x)$

$f(x) = cot(x) \Rightarrow f'(x) - cosec^2(x)$

$f(x) = arcsen \Rightarrow f'(x) = \frac{1}{\sqrt{1-x^2}}$

$f(x) = arccos \Rightarrow f'(x) = -\frac{1}{\sqrt{1-x^2}}$

$f(x) = arcTan(x) \Rightarrow f'(x) = \frac{1}{\cancel{1+x}}$